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🧪 Lab3 — MIME File Transfer over Router-on-a-Stick with Wireshark Analysis

A Cisco Networking Academy–style lab that blends classical networking with application-layer engineering. You will configure inter-VLAN routing, deploy a MIME-aware socket server, capture packets, and analyze how meaning flows through the stack.

📘 Lab Overview

Scenario

A client in one VLAN transfers files to a server in another VLAN using a custom MIME-based protocol (JSON header + binary payload). You will verify connectivity, observe encapsulation, and analyze packets using Wireshark.

This lab reinforces how Layer 2 segmentation, Layer 3 routing, TCP reliability, and application metadata interact.

🎯 Objectives

After completing this lab, you will be able to:

- Configure Router-on-a-Stick inter-VLAN routing
- Configure VLANs and trunking on a switch
- Deploy and run a MIME socket server and client
- Capture traffic using Wireshark
- Identify ARP, VLAN tags, TCP handshake, and application data
- Decode JSON headers in packet payload
- Troubleshoot network and application failures
- Document protocol behavior across OSI layers

🧰 Required Resources

- 1 Router (e.g., Cisco Systems ISR/virtual router)
- 1 Switch (Layer 2)
- 2 PCs or VMs
- Ethernet cables or virtual links
- Python 3 installed
- MIME file transfer project
- Wireshark installed (from Wireshark Foundation)

🗺️ Topology

```
None
PC1 (VLAN 10) ----\
                    \ access
                     \
                     [S1]---- trunk ----[R1]
                     /
                    / access
PC2 Server (VLAN 20)
```

📋 Addressing Scheme

Device	Interface	IP	Gateway
R1	G0/0.10	192.168.10.1/24	—
R1	G0/0.20	192.168.20.1/24	—
PC1-Client	NIC	192.168.10.10	192.168.10.1
PC2-Server	NIC	192.168.20.20	192.168.20.1

🧠 Background Theory

Router-on-a-Stick

A single physical interface carries multiple VLANs via 802.1Q tagging. Subinterfaces perform routing between broadcast domains.

MIME Metadata

MIME labels content type (e.g., image/png). Your protocol wraps files with JSON describing:

- MIME type
- Size

Why This Matters

Real systems — APIs, storage services, message brokers — depend on this exact pattern: structured metadata + binary payload over reliable transport.

🔧 Part 1 — Configure Network Devices

Step 1 — Router Configuration

```
None
enable
configure terminal
hostname R1

interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

end
write memory
```

Step 2 — Switch Configuration

```
None
enable
configure terminal
hostname S1

vlan 10
name CLIENTS

vlan 20
name SERVERS

interface g0/1
switchport mode trunk

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 20

end
write memory
```

Step 3 — Configure Hosts

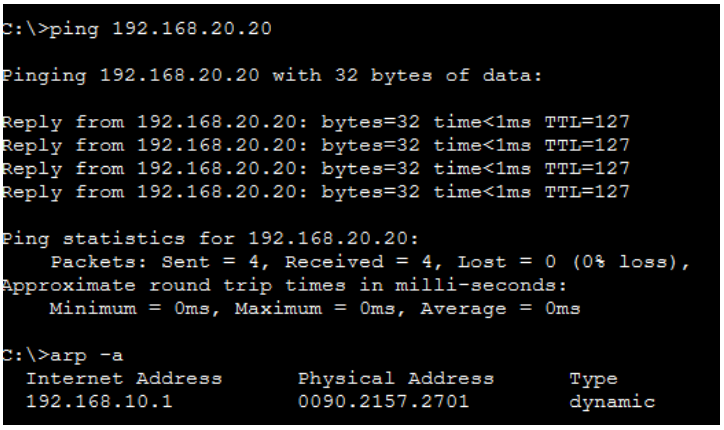
Set static IP addresses per table.

🔍 Part 2 — Verify Connectivity

From PC1:

```
None
ping 192.168.20.20
```

Verify:

- 

```
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0090.2157.2701        dynamic
```

Router checks:

```
None
show ip route
show ip interface brief
```

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20

R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Switch checks:

```
None
show interfaces trunk
show mac address-table
```

```
S1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20

S1>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0090.2157.2701    DYNAMIC     Gig0/1
  10    0090.2157.2701    DYNAMIC     Gig0/1
  20    0090.2157.2701    DYNAMIC     Gig0/1
```

🧪 Part 3 — Deploy MIME Server

On Server (PC2) (ทำผ่าน local host)

```
None
cd automation
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

```
PS D:\Code\week01-mime-typing-main> py server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
[20:06:37] INFO: Starting server with host=0.0.0.0, port=65432, storage=storage
[20:06:37] INFO: Server listening on 0.0.0.0:65432
```

On Client (PC1)

```
None
cd automation
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port
65432
```

```
PS D:\Code\week01-mime-typing-main> py client\main_enhanced.py assets\sample.png --host 127.0.0.1 --port 65432
[20:07:19] INFO: Starting client with 1 files to 127.0.0.1:65432
[20:07:19] INFO: Connected to 127.0.0.1:65432
[20:07:19] INFO: Sent assets\sample.png (108 bytes) as image/png
[20:07:19] INFO: Connection closed. Sent: 1, Failed: 0
[20:07:19] INFO: All files sent successfully
PS D:\Code\week01-mime-typing-main>
```

Confirm file saved.

📡 Part 4 — Wireshark Capture Walkthrough on PC1 and PC1

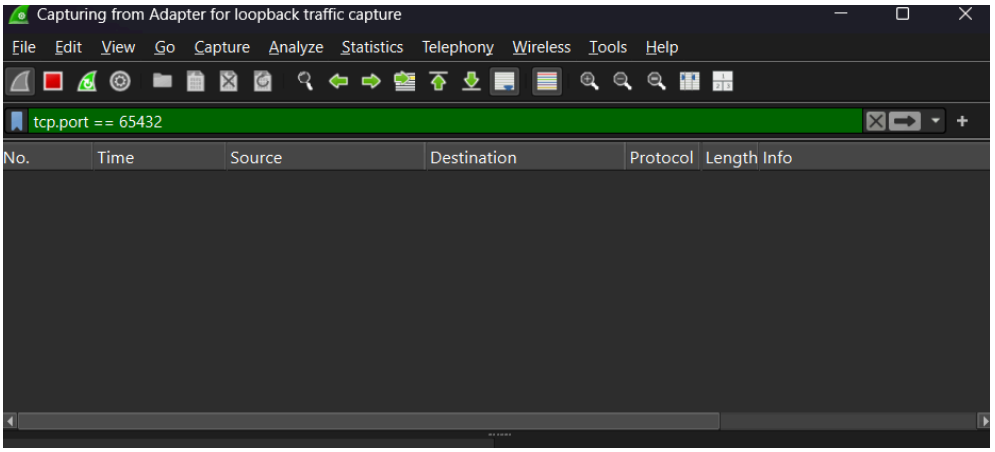
Step 1 — Start Capture

Open Wireshark.

Select interface connected to network.

Apply capture filter:

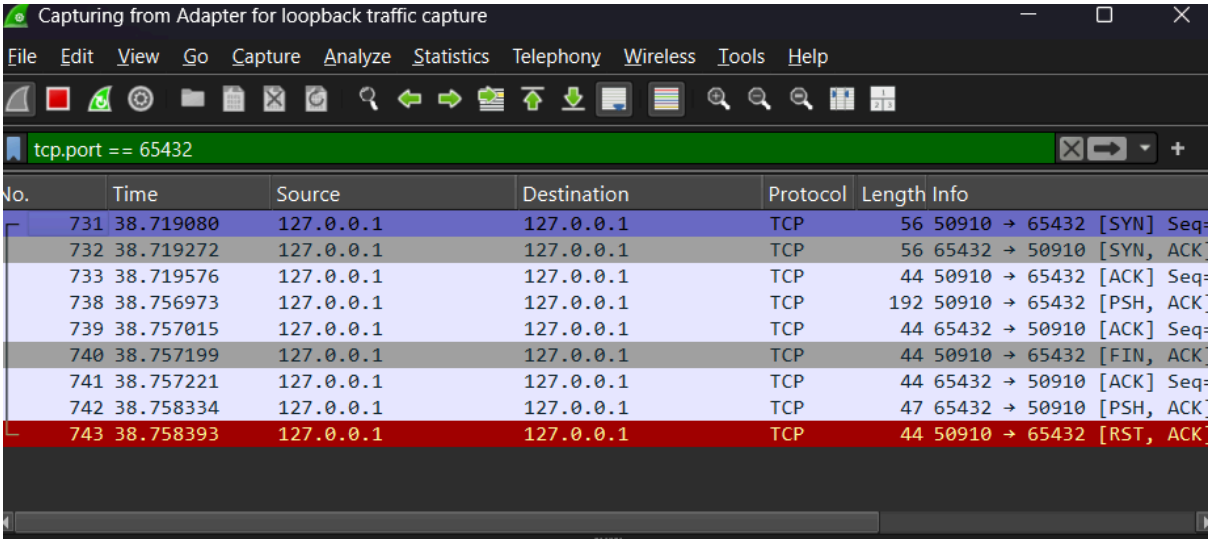
```
None
tcp port 65432
```



Step 2 — Initiate Transfer

Send file again.

Stop capture after completion.



Step 3 — Analyze Packets

A — ARP Exchange

Look for:

- Who has 192.168.20.20?
  ในการทดลองนี้ใช้ loopback address (127.0.0.1) แทนการส่งข้าม VLAN จริง ดังนั้นจึงไม่มี ARP exchange เกิดขึ้น เนื่องจาก ARP ทำงานที่ Layer 2 เพื่อหา MAC address ของ host ในเครือข่ายเดียวกัน แต่ loopback ไม่ผ่าน physical network interface จึงไม่จำเป็นต้องใช้ ARP

Confirm MAC resolution.

B — TCP Three-Way Handshake

Identify:

1 SYN 2 SYN-ACK 3 ACK

Check sequence numbers.

Length	Info
56	50910 → 65432 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SA…
56	65432 → 50910 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=6549…
44	50910 → 65432 [ACK] Seq=1 Ack=1 Win=65280 Len=0

C — Data Transfer

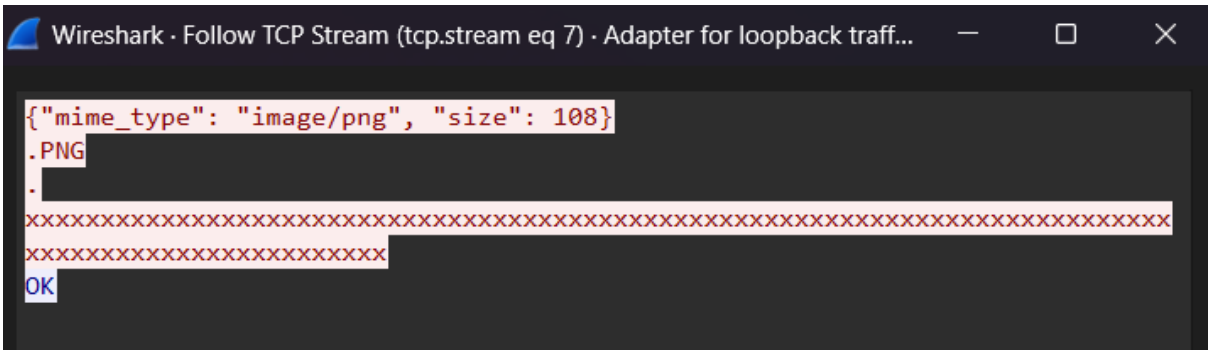
Follow stream:

Right click → Follow → TCP Stream.

You should see JSON header similar to:

```
None
{"mime_type":"image/png","size":12345}
```

Followed by binary.


Wireshark · Follow TCP Stream (tcp.stream eq 7) · Adapter for loopback traff...
{"mime_type": "image/png", "size": 108}
.PNG
.
xxxx

D — VLAN Tags

If capturing on trunk:

Look for:

```
None
802.1Q Virtual LAN
```

Note VLAN ID.
เนื่องจากการทดลองนี้ใช้ loopback address บนเครื่องเดียวกัน traffic จึงไม่ผ่าน physical switch ดังนั้นจึงไม่พบ 802.1Q VLAN tag ใน packet capture หากทดลองบน topology จริง packet ที่วิ่งบน trunk link ระหว่าง Switch S1 กับ Router R1 จะมี 802.1Q tag ระบุ VLAN ID 10 สำหรับ traffic จาก PC1 และ VLAN ID 20 สำหรับ traffic จาก PC2

🔬 Packet Dissection Guide

Layer	Field	What to Observe
Ethernet	Src/Dst MAC	Frame direction
802.1Q	VLAN ID	Segmentation
IP	TTL	Routing behavior
TCP	Flags	Reliability
Payload	JSON	MIME metadata

🛠️ Part 5 — Troubleshooting Labs

Scenario 1 — Broken Routing

Shutdown subinterface.

Observe:

```
PS D:\Code\week01-mime-typing-main> py client/main_enhanced.py assets/sample.png --host 127.0.0.1 --port 65432
[19:54:28] INFO: Starting client with 1 files to 127.0.0.1:65432
[19:54:30] ERROR: Failed to connect to 127.0.0.1:65432: [WinError 10061] No connection could be made because the target
machine actively refused it
[19:54:30] ERROR: Some files failed to send
```

- Ping failure
- TCP timeout

Scenario 2 — Firewall Block

Block port.

เนื่องจาก run ผ่าน local host จึง bypass firewall

Observe:

- SYN retransmissions

Scenario 3 — Wrong Gateway

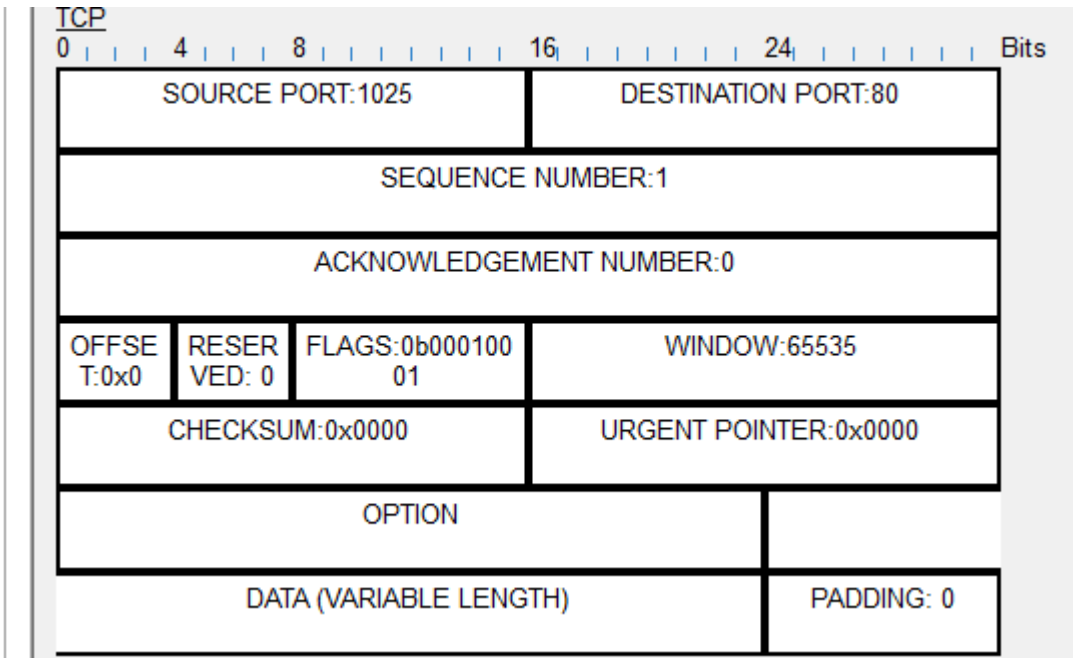
```
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>arp -a
No ARP Entries Found
C:\>
```

Observe:

- ARP incomplete
- (ทำใน packet tracer แก้ gateway เป็นค่า ที่ไม่มี, แก้ด้วยการเชื่อมด้วย gateway ที่มีอยู่ใน router)

Scenario 4 — Packet Loss Simulation

Drop packets.



Observe:

- TCP retransmissions
- Window adjustment

📝 Lab Report Requirements

Include:

- Screenshots of configurations
- Ping results
- Wireshark capture screenshots
- JSON header evidence
- Encapsulation explanation
- Troubleshooting notes

🎓 Assessment Questions

1. Why does TCP retransmit lost segments?
    - เพราะ TCP เป็น โปรโตคอลที่ รับประกันข้อมูลมากกว่า ความเร็วจึงทำการส่งซ้ำเพื่อไม่ให้ข้อมูลหาย
2. How does VLAN tagging affect broadcast domains?
    - VLAN tagging แบ่ง broadcast domain ขนาดใหญ่ออกเป็น domain ย่อยๆ
3. What role does MIME play compared to TCP headers?
    - MIME และ TCP headers ทำงานคนละชั้น (Layer) และมีหน้าที่ต่างกัน
4. Where does routing decision occur?
    - เกิดขึ้นที่ Layer 3 (Network Layer) ใน Router
5. Why does ARP precede TCP communication?
    - เพราะ TCP ทำงานที่ Layer 4 แต่ต้องส่งข้อมูลผ่าน Layer 2 (Ethernet) ซึ่งใช้ MAC address ในการส่งถึงกัน

📊 Reflection Exercise

Explain:

> How does application meaning survive multiple encapsulation layers?
> ความหมายของ application ยังคงอยู่เพราะ แต่ละ Layer จะห่อหุ้ม (encapsulate) ข้อมูลเดิมไว้ทั้งหมด เหมือนการใส่ซองจดหมายหลายชั้น แต่ข้างในยังคงเป็นจดหมายฉบับเดิม

⚠️ Common Pitfalls

- Server bound to localhost only
- Missing trunk
- Wrong VLAN
- OS firewall enabled
- Capture on wrong interface

🚀 Extension Activities

- Add TLS encryption
- Measure throughput vs latency
- Capture checksum errors
- Add checksum field to protocol
- Containerize server
- Compare with HTTP transfer

🧬 Instructor Notes — What This Lab Really Teaches

This is not just networking.

It demonstrates:

- Protocol layering philosophy
- Reliability engineering
- Metadata semantics
- Distributed system fundamentals

The student watches a file become:

> intention → bytes → frames → signals → bytes → meaning

That transformation is the essence of computer networks.